



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VETRI VELAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

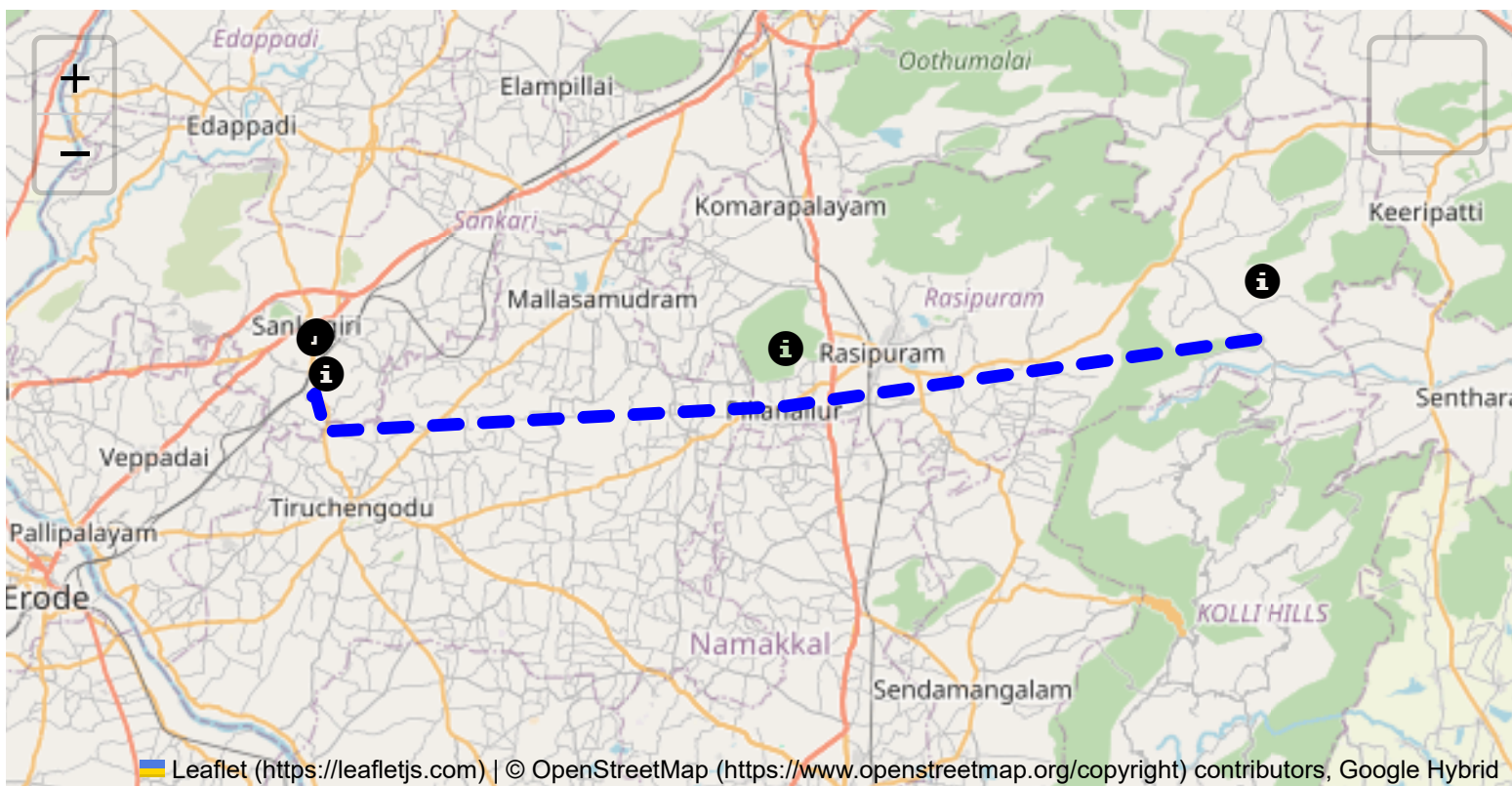
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

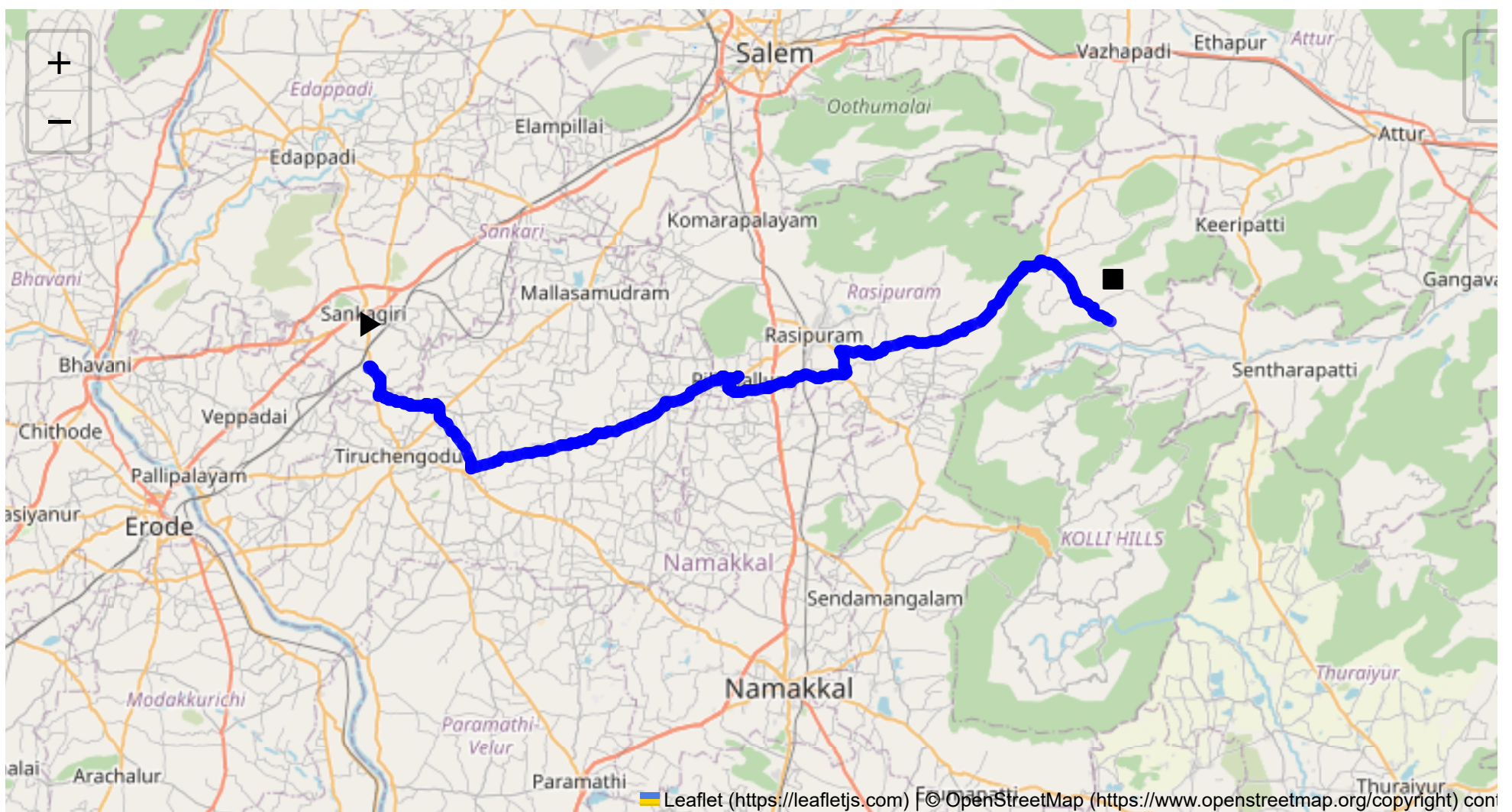
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 74.46 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours
Start: (11.4381, 77.8734)
End: (11.469806, 78.392995)



Welcome to the Journey Risk Management Study

Route Overview

The journey spans approximately 74.46 kilometers from Sangagiri to Bailnadu R.F. North, passing through Kangeyampalayam, Palmadai, and Gurusamipalayam. The route generally follows local roads and State Highway 86, linking various rural and semi-urban areas.

Weather Conditions

1. **Typical Weather:** Tamil Nadu experiences a tropical climate. The region has hot summers (March-May), monsoons (June-September), and mild winters (October-February).
2. **Potential Hazards:** During the monsoon, heavy rains can cause poor visibility and flooding, especially on rural roads that may not have advanced drainage systems.

Traffic Patterns

1. **Peak Hours:** Mornings (8 AM - 11 AM) and evenings (4 PM - 7 PM) see increased traffic due to local commuting and transport activities.
2. **Congestion-Prone Areas:** Small towns like Rasipuram can experience congestion due to narrow roads and market activity.

Road Quality and Infrastructure

1. **Road Conditions:** Varying road qualities; urban roads are mostly paved, while some sections through rural areas may have potholes or unpaved parts.
2. **Infrastructure Challenges:** Limited road signage, narrow bridges, and absence of lighting in rural stretches pose safety challenges.

Alternative Routes

In emergencies, alternative routes include:

- Detouring via nearby towns with better medical facilities or infrastructure.
- Avoiding smaller rural roads when possible and sticking to main highways (State or National).

Local Regulations on Hazardous Material Transport

1. **Permits:** Requires proper certification and documentation for transporting hazardous materials.
2. **Time Restrictions:** Transport may be restricted to specific hours to avoid peak traffic.

Historical Incidents

While specific incidents in this region may not be extensively documented, nearby industrial areas have reported isolated cases of hazardous material spills due to poor road conditions or human error.

Environmental Considerations

1. **Sensitive Areas:** Proximity to agricultural fields; ensure no contamination occurs in case of a spill.
2. **Wildlife:** Occasionally, the route may pass through areas where wildlife could cross, especially near forested regions.

Communication Coverage

1. **Coverage Analysis:** Major towns and highways generally have good mobile network coverage. Remote rural stretches may have signal drop-offs.
2. **Potential Dead Zones:** Expect communication breaks primarily in stretches far from towns or urban areas.

Emergency Response

1. **Major Towns:** Emergency services are more responsive around Sangagiri and Rasipuram due to better facilities and accessibility.
2. **Remote Areas:** Increased response times, sometimes over an hour, due to distance and terrain.

Overall Summary of Risk Assessment

- **Moderate to High Risk** during monsoon due to weather challenges.
- **Medium Risk** of road accidents, particularly in rural areas with poor road conditions.
- Safety improvements should focus on enhanced communication systems, driver training on weather adaptations, and better road maintenance.

This route requires careful planning, ongoing weather monitoring, and regular communication to minimize transportation risks for hazardous materials. It's essential to brief drivers on local conditions and equip them with necessary safety and communication tools.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44022, 77.87546	15 KM/Hr
2	Turn	High	11.41978, 77.88077	15 KM/Hr
3	Turn	Medium	11.41876, 77.88487	30 KM/Hr
4	Turn	Medium	11.41836, 77.88540	30 KM/Hr
5	Turn	Medium	11.41834, 77.88585	30 KM/Hr
6	Turn	Medium	11.41849, 77.88613	30 KM/Hr
7	Turn	High	11.41852, 77.88700	15 KM/Hr
8	Turn	Medium	11.41769, 77.88730	30 KM/Hr
9	Turn	Medium	11.41721, 77.88912	30 KM/Hr
10	Turn	Medium	11.41606, 77.89452	30 KM/Hr
11	Turn	Medium	11.41430, 77.90034	30 KM/Hr
12	Turn	Medium	11.41385, 77.90064	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
13	Turn	Medium	11.41374, 77.90097	30 KM/Hr
14	Turn	Medium	11.41388, 77.90175	30 KM/Hr
15	Turn	High	11.41235, 77.90492	15 KM/Hr
16	Turn	High	11.41271, 77.90516	15 KM/Hr
17	Turn	Medium	11.41292, 77.91162	30 KM/Hr
18	Turn	High	11.41418, 77.91421	15 KM/Hr
19	Turn	High	11.41409, 77.91458	15 KM/Hr
20	Turn	High	11.41205, 77.91566	15 KM/Hr
21	Turn	Medium	11.41216, 77.91668	30 KM/Hr
22	Turn	Medium	11.41188, 77.91733	30 KM/Hr
23	Turn	Medium	11.41347, 77.92096	30 KM/Hr
24	Turn	Medium	11.41336, 77.92130	30 KM/Hr
25	Turn	Medium	11.40862, 77.92362	30 KM/Hr
26	Turn	Medium	11.40681, 77.92309	30 KM/Hr
27	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
28	Turn	Medium	11.41321, 78.08067	30 KM/Hr
29	Turn	High	11.41453, 78.08062	15 KM/Hr
30	Turn	Medium	11.41476, 78.08094	30 KM/Hr
31	Turn	Medium	11.43240, 78.12064	30 KM/Hr
32	Turn	Medium	11.43258, 78.12110	30 KM/Hr
33	Blind Spot	Blind Spot	11.43260, 78.13252	10 KM/Hr
34	Turn	High	11.43131, 78.12710	15 KM/Hr
35	Turn	Medium	11.42708, 78.12675	30 KM/Hr
36	Turn	Medium	11.42597, 78.12536	30 KM/Hr
37	Turn	High	11.42447, 78.12483	15 KM/Hr
38	Turn	Medium	11.42450, 78.15278	30 KM/Hr
39	Turn	High	11.43542, 78.20754	15 KM/Hr
40	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
41	Turn	High	11.44941, 78.22277	15 KM/Hr
42	Turn	Medium	11.44806, 78.22308	30 KM/Hr
43	Turn	Medium	11.44783, 78.22334	30 KM/Hr
44	Turn	Medium	11.44753, 78.22458	30 KM/Hr
45	Turn	Medium	11.45366, 78.23608	30 KM/Hr
46	Turn	Medium	11.49639, 78.32346	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
47	Turn	Medium	11.50762, 78.33281	30 KM/Hr
48	Turn	High	11.51183, 78.34416	15 KM/Hr
49	Turn	Medium	11.51037, 78.34883	30 KM/Hr
50	Turn	Medium	11.48656, 78.36901	30 KM/Hr
51	Turn	Medium	11.47748, 78.38117	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
4	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
1	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
2	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
3	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44022, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41978, 77.88077



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41836, 77.88540



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41834, 77.88585



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41849, 77.88613



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41852, 77.88700



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41769, 77.88730



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41721, 77.88912



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41606, 77.89452



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41430, 77.90034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41385, 77.90064



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41374, 77.90097



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41388, 77.90175



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41235, 77.90492



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41271, 77.90516



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41292, 77.91162



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41418, 77.91421



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41409, 77.91458



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41205, 77.91566



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41216, 77.91668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41188, 77.91733



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41347, 77.92096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41336, 77.92130



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.40862, 77.92362



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40681, 77.92309



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.36920, 77.94568



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41321, 78.08067



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41453, 78.08062



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41476, 78.08094



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43240, 78.12064



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.43258, 78.12110



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.43260, 78.13252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43131, 78.12710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42708, 78.12675



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42597, 78.12536



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.42447, 78.12483



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.42450, 78.15278



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.43542, 78.20754



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.45100, 78.20385



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44941, 78.22277



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44806, 78.22308



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44783, 78.22334



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44753, 78.22458



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45366, 78.23608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49639, 78.32346

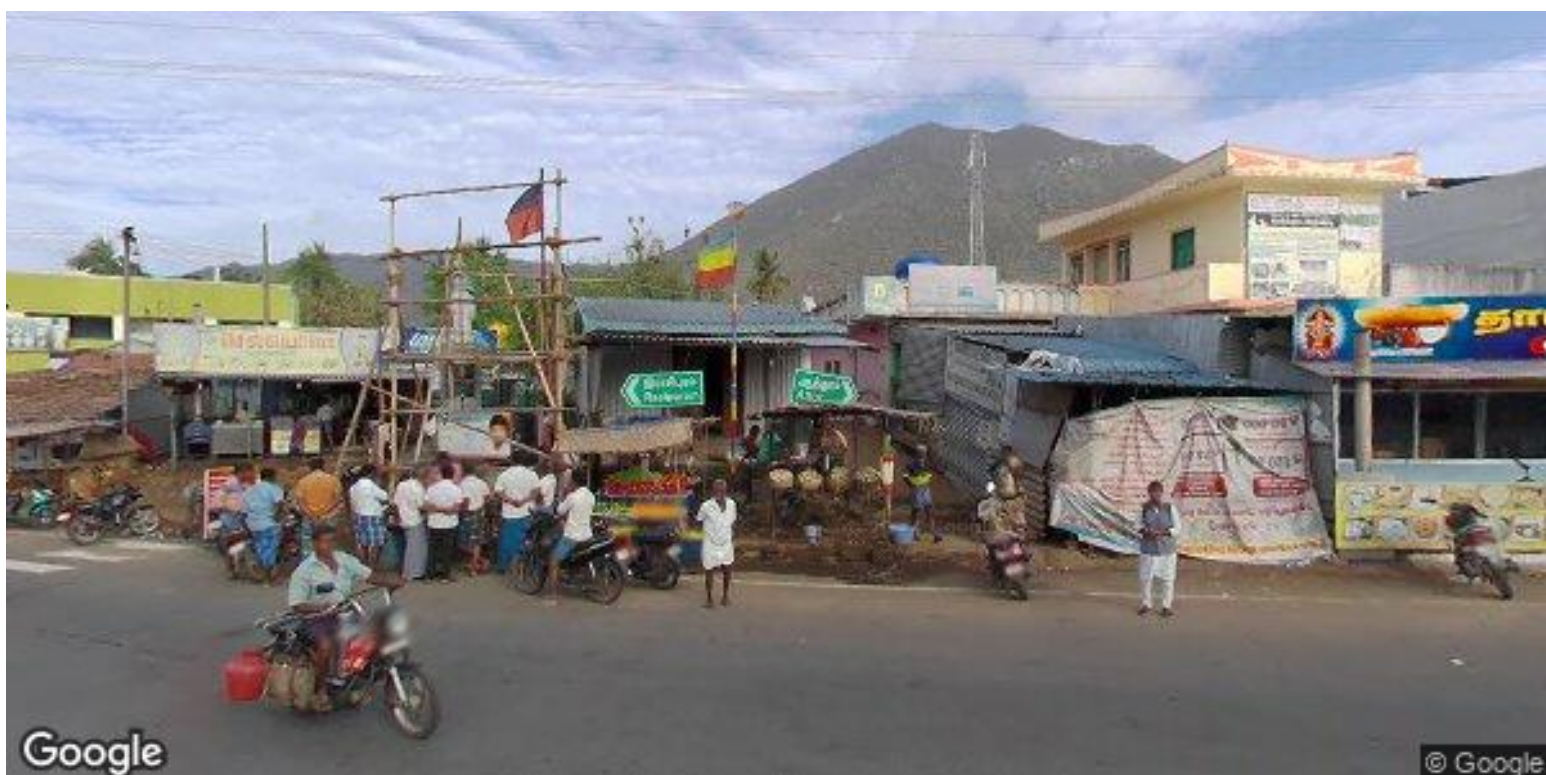


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50762, 78.33281



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51037, 78.34883



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47748, 78.38117
